



JKBOSE

🔥 2026 CLASS 11th 🔥

SETS

MATHS

LECTURE-01

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Introduction to Sets

2

Representation of Sets

3

Types of Sets

4

Operation on Sets and Venn Diagram

5

Related problems

14 chapters

syllabus:

Unit I $\longrightarrow$ Sets and functions ^③
 $\swarrow$ Sets
 $\searrow$ Relations & functions
 $\searrow$ Trigonometric functions

Unit II $\longrightarrow$ Algebra ^⑤

Unit III $\longrightarrow$ Co-ordinate Geometry — st. lines, Conic Sections, 3D. ^③

Unit IV $\longrightarrow$ Calculus ^① $\longrightarrow$ Limits & Derivatives.

Unit V $\longrightarrow$ statistics & probability ^②

Introduction

* What is a Set?

Definition:

- A set is a **well defined** collection of objects.

↓
(Same meaning
for everyone)

* Collection of **Good Actors** in Bollywood.
↓
It is not a set.

• Collection of **Beautiful Girls** in an Area. (It is not a set)

• Collection of Natural numbers
(It is a set)

• Collection of Handsome Boys (not a set)

• Collection of Good Teachers in Physics Wallah (not a set)

EARLY 400.

key points:

- well defined $\Rightarrow$ There should be no Ambiguity
- Distinct $\Rightarrow$ Repetition of objects is not Allowed.
- members of the set are known as the elements of the set.
- The number of elements in a Set is known as the Cardinality of the set.
- Sets are Denoted by Capital letters.

Question

A Set is a

- ~~1~~ Collection of Objects ✗
- ~~2~~ Collection of Numbers ✗
- ~~3~~ Well defined Collection of Objects ✓
- 4 Well defined Collection of Numbers

N : Set of natural numbers.

W : Set of whole numbers.

Z : Set of Integers

Q : Set of Rational numbers.

R : Set of Real numbers.

Z^+ : Set of Positive Integers

Q^+ : Set of positive rational numbers.

R^+ : Set of Positive real numbers.

Question

The set of Rational numbers is Usually denoted by

1 Q ✓

2 R ✗

3 Z ✗

4 N ✗

Question

Which of the following Collections is not a Set?

- 1 The collection of Natural numbers between 1 and 20 ✗
- 2 The collection of Roots of the equation $x^2 - 5x + 6 = 0$ ✗
 $(x-2)(x-3)=0$ $x=2,3$
- 3 The collection of Prime numbers between 1 and 20 ✗
- 4 The collection of Intelligent Students of J & K ✓✓

Question

If A denotes the **set of Prime numbers between 1 and 30**, then the cardinality of A is $n(A)$

1 8 ✗

2 10 ✓

3 9 ✗

4 11 ✗

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

Question

If B denotes the **set of Vowels in English Alphabets** then the **cardinality of B** is equal to

☒ 1 5

☒ 2 4

☒ 3 21

☒ 4 8

Jon Mehra 1729 (no of vowels)

a, e, i, o, u
#

Thank
You